

A Fixed-Center Accelerated Aggregate-Cut Method for Finite-Diameter Hybrid Convex Composite Optimization

Partial Progress Toward the Accelerated GPB Question of Liang–Monteiro

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Abstract

Liang and Monteiro asked whether one can design an accelerated variant of their generic proximal bundle (GPB) framework that is optimal for the finite-diameter hybrid convex composite optimization (HCCO) class. This note proves a closely related, but not identical, result. We give a fixed-center accelerated aggregate-cut method for

$$\min_{x \in \mathbb{R}^n} \phi(x) := f(x) + h(x),$$

where f, h are proper closed convex functions, $\text{dom } h \subseteq \text{dom } f$, the set $Q := \text{dom } h$ has finite diameter D , and a first-order oracle for f satisfies

$$\|f'(u) - f'(v)\| \leq 2M_f + L_f \|u - v\|, \quad u, v \in Q.$$

The method stores affine cuts of f , keeps h exact, and solves one regularized aggregate-cut subproblem per iteration. With exact subproblem solutions and known M_f, L_f, D , it obtains

$$\phi(x_N) - \phi^* \leq \frac{2L_f D^2}{N^2} + \frac{4M_f D}{\sqrt{N}},$$

and hence has oracle complexity

$$O\left(\frac{\sqrt{L_f} D}{\sqrt{\varepsilon}} + \frac{M_f^2 D^2}{\varepsilon^2} + 1\right).$$

This matches the optimal finite-diameter HCCO rate cited by Liang–Monteiro. However, the method is not, as stated, a GPB variant in their sense: it has a fixed prox-center, variable aggregate weights, and no serious/null iteration cycle. Thus the result should be read as partial progress toward the accelerated GPB problem, not as a complete solution of that problem.

1 Status of the result

The mathematical result below is a valid rate theorem for an accelerated proximalized aggregate-cut scheme. It is not a full answer to the GPB-specific open question in [1, Section 6]. The distinction is important. Liang and Monteiro’s GPB framework is organized around current prox-centers, bundle update schemes, and serious/null iterations. The algorithm analyzed here is instead a Nesterov estimate-sequence or dual-averaging style method with a fixed center x_0 and a weighted aggregate of cuts.

Consequently, this note establishes the following partial progress.

There is a clean aggregate-cut proximal method that attains the finite-diameter optimal HCCO rate. The proof does not show that Liang–Monteiro’s GPB framework itself can be accelerated to the same rate.

No parameter-free or adaptive implementation is proved here. The horizon N and curvature parameter used by the method are chosen from M_f, L_f, D and the target accuracy. Exact solution of the regularized aggregate subproblem is also assumed, as is common in oracle-complexity statements for bundle methods.

2 Problem class

Let $f, h : \mathbb{R}^n \rightarrow \mathbb{R} \cup \{+\infty\}$ be proper closed convex functions and assume

$$Q := \text{dom } h \subseteq \text{dom } f, \quad D := \sup\{\|x - y\| : x, y \in Q\} < \infty.$$

Since h is convex, Q is convex. We assume Q is nonempty and choose an initial point $x_0 \in Q$. Let

$$\phi(x) := f(x) + h(x), \quad \phi^* := \inf_{x \in \mathbb{R}^n} \phi(x) = \inf_{x \in Q} \phi(x).$$

A first-order oracle queried at $y \in Q$ returns $f(y)$ and some subgradient $f'(y) \in \partial f(y)$. The HCCO hybrid condition is that, for all $u, v \in Q$,

$$\|f'(u) - f'(v)\| \leq 2M_f + L_f \|u - v\|, \quad (1)$$

where $M_f, L_f \geq 0$.

For $y \in Q$ define the affine cut

$$\ell_y(x) := f(y) + \langle f'(y), x - y \rangle. \quad (2)$$

Convexity gives $\ell_y(x) \leq f(x)$ for all $x \in Q$.

3 A hybrid upper model

The following elementary lemma converts (1) into the upper model needed for acceleration.

Lemma 1 (hybrid quadratic-linear model). *For every $x, y \in Q$,*

$$f(x) \leq \ell_y(x) + 2M_f \|x - y\| + \frac{L_f}{2} \|x - y\|^2. \quad (3)$$

Consequently, for every $\eta > 0$,

$$f(x) \leq \ell_y(x) + \frac{L_f + \eta}{2} \|x - y\|^2 + \frac{2M_f^2}{\eta}. \quad (4)$$

If $M_f = 0$, the same statement holds with $\eta = 0$ and zero additive error.

Proof. The lower-cut property follows from the subgradient inequality; we prove the upper bound. Fix $x, y \in Q$ and write $d = x - y$. If $d = 0$, there is nothing to prove. Otherwise, for an integer $p \geq 1$, set

$$z_i := y + \frac{i}{p}d, \quad i = 0, 1, \dots, p.$$

The points z_i lie in Q by convexity. Since $f'(z_i) \in \partial f(z_i)$,

$$f(z_i) - f(z_{i-1}) \leq \langle f'(z_i), z_i - z_{i-1} \rangle = \frac{1}{p} \langle f'(z_i), d \rangle.$$

Summing over $i = 1, \dots, p$ gives

$$\begin{aligned} f(x) - f(y) &\leq \frac{1}{p} \sum_{i=1}^p \langle f'(z_i), d \rangle \\ &= \langle f'(y), d \rangle + \frac{1}{p} \sum_{i=1}^p \langle f'(z_i) - f'(y), d \rangle \\ &\leq \langle f'(y), d \rangle + \frac{\|d\|}{p} \sum_{i=1}^p (2M_f + L_f \|z_i - y\|) \\ &\leq \langle f'(y), d \rangle + 2M_f \|d\| + \frac{L_f \|d\|^2}{p} \sum_{i=1}^p \frac{i}{p} \\ &= \langle f'(y), d \rangle + 2M_f \|d\| + \frac{L_f}{2} \left(1 + \frac{1}{p}\right) \|d\|^2. \end{aligned}$$

Letting $p \rightarrow \infty$ yields (3). Finally,

$$2M_f r \leq \frac{\eta}{2} r^2 + \frac{2M_f^2}{\eta}, \quad r \geq 0,$$

which proves (4). If $M_f = 0$, the linear term in (3) is absent, so one may take $\eta = 0$. $\square$

4 The accelerated aggregate-cut method

Fix a horizon $N \geq 1$ and a curvature parameter $H > 0$. Define a sequence $(A_k)_{k \geq 0}$ by $A_0 = 0$ and, at iteration k , choose $a_{k+1} > 0$ so that

$$H a_{k+1}^2 = A_{k+1} := A_k + a_{k+1}. \tag{5}$$

Equivalently,

$$a_{k+1} = \frac{1 + \sqrt{1 + 4HA_k}}{2H}.$$

Set $\alpha_{k+1} := a_{k+1}/A_{k+1}$.

Algorithm 1: Fixed-center accelerated aggregate-cut method.

1. Choose $x_0 \in Q$. Set $v_0 = x_0$, $A_0 = 0$, and

$$\Psi_0(x) := \frac{1}{2} \|x - x_0\|^2.$$

2. For $k = 0, 1, \dots, N - 1$:

- (a) Choose a_{k+1} by (5), set $A_{k+1} = A_k + a_{k+1}$ and $\alpha_{k+1} = a_{k+1}/A_{k+1}$.

(b) Define the extrapolated oracle point

$$y_k = (1 - \alpha_{k+1})x_k + \alpha_{k+1}v_k. \quad (6)$$

Query the oracle at y_k and form the cut

$$\ell_k(x) := f(y_k) + \langle f'(y_k), x - y_k \rangle. \quad (7)$$

(c) Update

$$\Psi_{k+1}(x) := \Psi_k(x) + a_{k+1}(\ell_k(x) + h(x)) \quad (8)$$

and compute

$$v_{k+1} := \operatorname{argmin}_{x \in \mathbb{R}^n} \Psi_{k+1}(x). \quad (9)$$

(d) Set

$$x_{k+1} := (1 - \alpha_{k+1})x_k + \alpha_{k+1}v_{k+1}. \quad (10)$$

3. Return x_N .

The minimizer in (9) exists and is unique because Ψ_{k+1} is proper closed and 1-strongly convex. Since $h = +\infty$ outside Q , all v_k and x_k lie in Q .

Remark 1 (aggregate-cut subproblem). *For $k \geq 1$, define the weighted aggregate cut*

$$\bar{\ell}_k(x) := \frac{1}{A_k} \sum_{i=0}^{k-1} a_{i+1} \ell_i(x).$$

Since each ℓ_i is a global lower cut of f on Q , so is $\bar{\ell}_k$. The subproblem (9) is equivalent to

$$v_k = \operatorname{argmin}_x \left\{ \bar{\ell}_k(x) + h(x) + \frac{1}{2A_k} \|x - x_0\|^2 \right\}. \quad (11)$$

Thus the algorithm may store all weighted cuts or only their aggregate. This is an aggregate-cut model, not the max-of-cuts model used in many classical bundle methods.

5 Estimate-sequence inequality

We first prove a deterministic bound that only assumes a quadratic upper model with additive error.

Lemma 2 (accelerated aggregate-cut bound). *Assume that for some $H > 0$ and $\delta \geq 0$, every oracle point $y \in Q$ satisfies*

$$f(x) \leq \ell_y(x) + \frac{H}{2} \|x - y\|^2 + \delta, \quad x \in Q. \quad (12)$$

Then the iterates of Algorithm 1 satisfy, for every $u \in Q$,

$$\phi(x_N) - \phi(u) \leq \frac{\|u - x_0\|^2}{2A_N} + \delta \frac{\sum_{i=1}^N A_i}{A_N}. \quad (13)$$

Moreover,

$$A_N \geq \frac{N^2}{4H}, \quad (14)$$

so

$$\phi(x_N) - \phi(u) \leq \frac{2H \|u - x_0\|^2}{N^2} + N\delta. \quad (15)$$

Proof. Let

$$E_k := \delta \sum_{i=1}^k A_i, \quad E_0 := 0.$$

We prove by induction that

$$A_k \phi(x_k) \leq \Psi_k(v_k) + E_k, \quad k = 0, 1, \dots, N. \quad (16)$$

For $k = 0$ the statement is trivial because $A_0 = E_0 = 0$. Suppose (16) holds for some $k < N$. To simplify notation, write

$$a = a_{k+1}, \quad A = A_{k+1}, \quad \alpha = a/A.$$

From (6) and (10),

$$x_{k+1} - y_k = \alpha(v_{k+1} - v_k) = \frac{a}{A}(v_{k+1} - v_k). \quad (17)$$

Using (12) at $y = y_k$ and $x = x_{k+1}$,

$$A\phi(x_{k+1}) \leq A\ell_k(x_{k+1}) + Ah(x_{k+1}) + \frac{AH}{2} \|x_{k+1} - y_k\|^2 + A\delta. \quad (18)$$

Because ℓ_k is affine and h is convex,

$$\begin{aligned} A\ell_k(x_{k+1}) &= A_k\ell_k(x_k) + a\ell_k(v_{k+1}), \\ Ah(x_{k+1}) &\leq A_k h(x_k) + ah(v_{k+1}). \end{aligned}$$

Since $\ell_k(x_k) \leq f(x_k)$, (18) gives

$$A\phi(x_{k+1}) \leq A_k\phi(x_k) + a(\ell_k(v_{k+1}) + h(v_{k+1})) + \frac{AH}{2} \|x_{k+1} - y_k\|^2 + A\delta. \quad (19)$$

By the induction hypothesis, (17), and the weight identity $Ha^2 = A$,

$$\begin{aligned} A\phi(x_{k+1}) &\leq \Psi_k(v_k) + E_k + a(\ell_k(v_{k+1}) + h(v_{k+1})) + \frac{AH}{2} \left(\frac{a}{A}\right)^2 \|v_{k+1} - v_k\|^2 + A\delta \\ &= \Psi_k(v_k) + E_k + a(\ell_k(v_{k+1}) + h(v_{k+1})) + \frac{1}{2} \|v_{k+1} - v_k\|^2 + A\delta. \end{aligned}$$

The function Ψ_k is 1-strongly convex and minimized by v_k , hence

$$\Psi_k(v_{k+1}) \geq \Psi_k(v_k) + \frac{1}{2} \|v_{k+1} - v_k\|^2. \quad (20)$$

Combining the last two displays and using (8) yields

$$A\phi(x_{k+1}) \leq \Psi_{k+1}(v_{k+1}) + E_k + A\delta = \Psi_{k+1}(v_{k+1}) + E_{k+1}.$$

Thus (16) holds at $k+1$.

Now fix $u \in Q$. Since every cut is below f on Q ,

$$\Psi_N(u) = \frac{1}{2} \|u - x_0\|^2 + \sum_{i=0}^{N-1} a_{i+1}(\ell_i(u) + h(u)) \leq \frac{1}{2} \|u - x_0\|^2 + A_N\phi(u). \quad (21)$$

Since v_N minimizes Ψ_N , (16) and (21) give

$$A_N\phi(x_N) \leq \Psi_N(v_N) + E_N \leq \Psi_N(u) + E_N \leq \frac{1}{2} \|u - x_0\|^2 + A_N\phi(u) + E_N.$$

Dividing by A_N proves (13).

It remains to prove (14). Define $s_k := \sqrt{HA_k}$. From $Ha_{k+1}^2 = A_{k+1}$ and $A_{k+1} = A_k + a_{k+1}$,

$$s_{k+1}^2 - s_k^2 = s_{k+1}.$$

Thus

$$s_{k+1} - s_k = \frac{s_{k+1}}{s_{k+1} + s_k} \geq \frac{1}{2},$$

because $0 \leq s_k \leq s_{k+1}$. Since $s_0 = 0$, we have $s_N \geq N/2$, which is exactly (14). Finally, $A_i \leq A_N$ for $i \leq N$, so $\sum_{i=1}^N A_i \leq NA_N$; together with (14), this gives (15). $\square$

6 Complexity for the HCCO class

We now choose the curvature H and model error δ by smoothing the linear term in Lemma 1.

Theorem 1 (finite-diameter HCCO rate). *Assume $D > 0$ and $M_f + L_f > 0$. Fix $N \geq 1$. If $M_f > 0$, set*

$$\eta_N := \frac{M_f N^{3/2}}{D}, \quad H_N := L_f + \eta_N, \quad \delta_N := \frac{2M_f^2}{\eta_N}. \quad (22)$$

If $M_f = 0$, set

$$\eta_N := 0, \quad H_N := L_f, \quad \delta_N := 0. \quad (23)$$

Run Algorithm 1 for N iterations with $H = H_N$. Then

$$\phi(x_N) - \phi^* \leq \frac{2L_f D^2}{N^2} + \frac{4M_f D}{\sqrt{N}}. \quad (24)$$

Consequently, for any $\bar{\varepsilon} > 0$, it is enough to choose

$$N \geq \max \left\{ \frac{2\sqrt{L_f} D}{\sqrt{\bar{\varepsilon}}}, \frac{64M_f^2 D^2}{\bar{\varepsilon}^2} \right\} \quad (25)$$

iterations to guarantee $\phi(x_N) - \phi^ \leq \bar{\varepsilon}$. In particular, the oracle complexity is*

$$O \left(\frac{\sqrt{L_f} D}{\sqrt{\bar{\varepsilon}}} + \frac{M_f^2 D^2}{\bar{\varepsilon}^2} + 1 \right). \quad (26)$$

Proof. First suppose $M_f > 0$. Lemma 1 shows that (12) holds with $H = H_N$ and $\delta = \delta_N$. Lemma 2 gives, for every $u \in Q$,

$$\phi(x_N) - \phi(u) \leq \frac{2H_N \|u - x_0\|^2}{N^2} + N\delta_N.$$

Since both u and x_0 belong to Q , $\|u - x_0\| \leq D$. Hence

$$\begin{aligned} \phi(x_N) - \phi(u) &\leq \frac{2(L_f + \eta_N)D^2}{N^2} + \frac{2NM_f^2}{\eta_N} \\ &= \frac{2L_f D^2}{N^2} + \frac{2\eta_N D^2}{N^2} + \frac{2NM_f^2}{\eta_N} \\ &= \frac{2L_f D^2}{N^2} + \frac{2M_f D}{\sqrt{N}} + \frac{2M_f D}{\sqrt{N}} \\ &= \frac{2L_f D^2}{N^2} + \frac{4M_f D}{\sqrt{N}}. \end{aligned}$$

Taking the infimum over $u \in Q$ proves (24).

If $M_f = 0$, Lemma 1 gives (12) with $H = H_N = L_f$ and $\delta = 0$. The same argument yields

$$\phi(x_N) - \phi^* \leq \frac{2L_f D^2}{N^2},$$

which is (24) with $M_f = 0$.

Finally, if (25) holds, then

$$\frac{2L_f D^2}{N^2} \leq \frac{\bar{\varepsilon}}{2}, \quad \frac{4M_f D}{\sqrt{N}} \leq \frac{\bar{\varepsilon}}{2},$$

so $\phi(x_N) - \phi^* \leq \bar{\varepsilon}$. This proves (26). $\square$

Corollary 1 (degenerate cases). *If $D = 0$, then Q contains at most one point and every feasible point is optimal. If $D > 0$ and $M_f = L_f = 0$, then the selected oracle subgradient is constant on Q , so f is affine on Q ; one exact aggregate-cut subproblem with a single cut minimizes $f + h$ exactly. Thus the complexity statement remains valid after the harmless addition of the $+1$ term in (26).*

Proof. The case $D = 0$ is immediate. If $M_f = L_f = 0$, then (1) gives $f'(u) = f'(v) =: g$ for all $u, v \in Q$. For any $u, v \in Q$, the subgradient inequalities

$$f(u) \geq f(v) + \langle g, u - v \rangle, \quad f(v) \geq f(u) + \langle g, v - u \rangle$$

force equality. Thus every cut is exact on Q . Minimizing $\ell_{x_0} + h = f + h$ over Q solves the problem. $\square$

7 Relation to the Liang–Monteiro question

The rate in Theorem 1 matches the finite-diameter optimal HCCO complexity quoted by Liang and Monteiro from Lan’s accelerated composite subgradient method [1, Section 6], [2]. The proof is short because the hybrid condition is used only through Lemma 1, and acceleration is handled entirely by the estimate-sequence inequality in Lemma 2.

However, this should not be presented as a complete solution to Liang–Monteiro’s GPB acceleration problem. The reasons are structural.

1. The algorithm has a fixed prox-center x_0 and a varying regularization coefficient $1/A_k$ in the aggregate subproblem (11).
2. It has no serious/null iteration mechanism and no GPB cycle termination test.
3. Its model is a weighted aggregate of cuts, rather than a GPB bundle function updated inside serious/null cycles.
4. The proof does not show how to accelerate a GPB method while preserving the GPB architecture of [1].

Thus the most accurate interpretation is the following. The note proves that the optimal finite-diameter HCCO rate is attainable by a proximalized aggregate-cut method, which is a bundle-like method in a broad modeling sense. It does not prove that the GPB framework itself admits such an acceleration. It is therefore genuine partial progress, and possibly a useful guide for constructing an accelerated GPB variant, but not a full resolution of the authors’ stated problem.

8 What is not proved here

Several potentially useful extensions are left open.

1. **A true accelerated GPB variant.** One would need to incorporate acceleration into a method with prox-center updates, serious/null iterations, and a GPB-compatible bundle update rule.
2. **Adaptive parameter selection.** The proof assumes knowledge of M_f, L_f, D through the choice of H_N and N . A logarithmic grid, restart, or line-search scheme may remove this assumption, but such a scheme is not analyzed here.
3. **Inexact subproblem solution.** The proof assumes that (9) is solved exactly. A practical bundle implementation would require an error-tolerant version of Lemma 2.
4. **Lower-bound certificates.** The finite-horizon theorem gives an a priori guarantee; it does not by itself produce a stopping certificate based only on observed cuts.

References

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